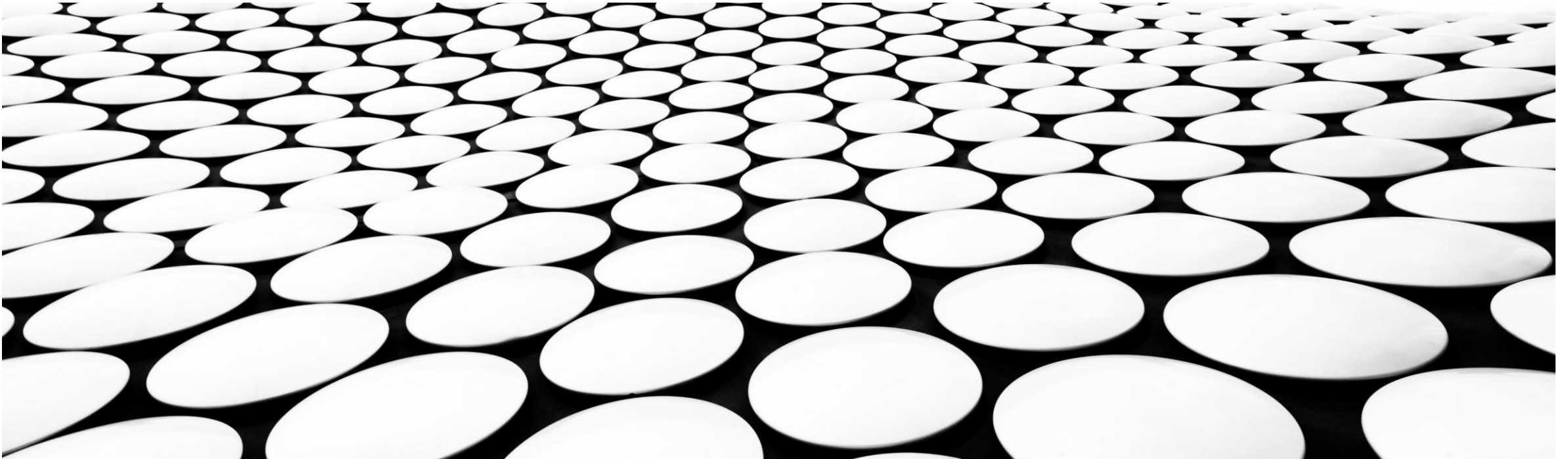
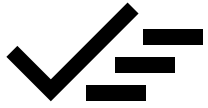


LINEAR REGRESSION

Examples, model learning, assumptions





LINEAR REGRESSION

- Introduction / Examples
- Learning model parameters
- Underlying assumptions
- Back on regression imputation

Look into Linear
Regression

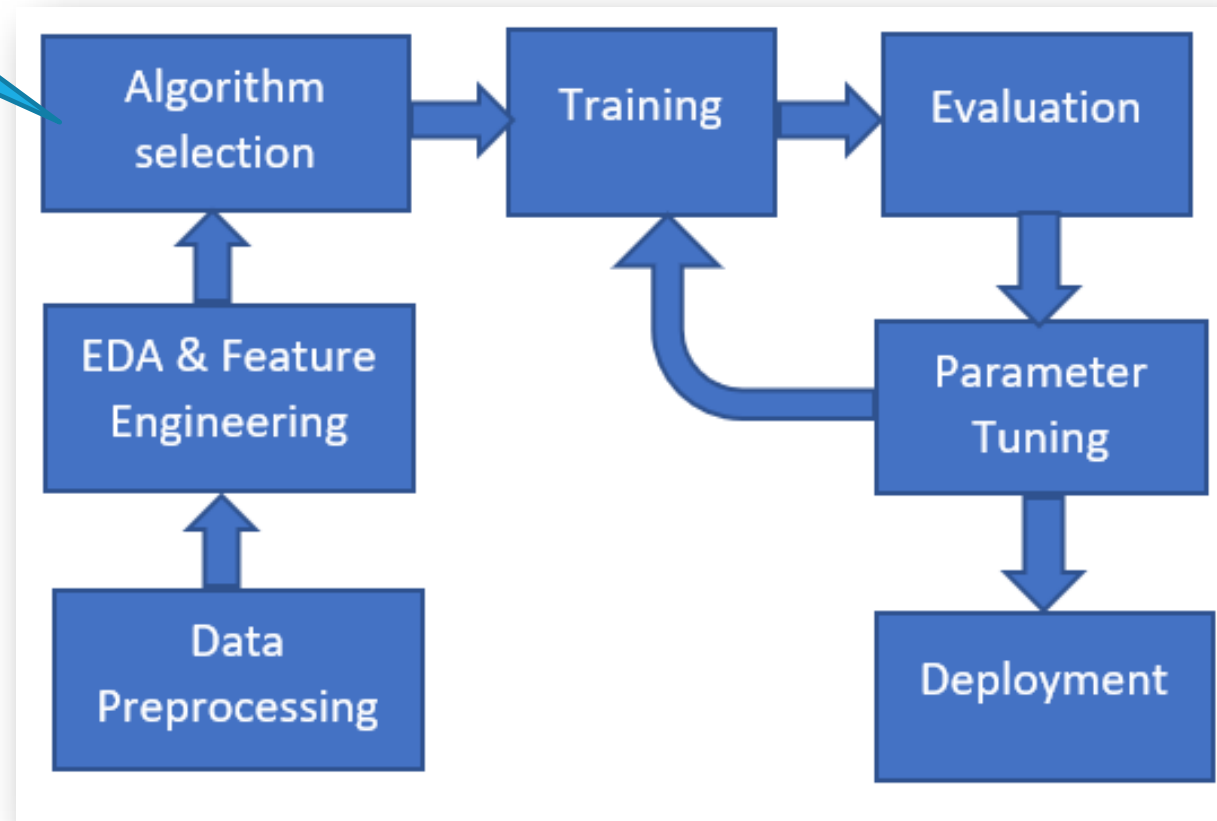
Classification

Naive Bayes
Decision Trees

Regression

Linear Regression

Supervised Machine Learning



Linear regression

Introduction / Examples



[Source](#)

$$\hat{y} = b \cdot x + a$$

Estimated dependent variable

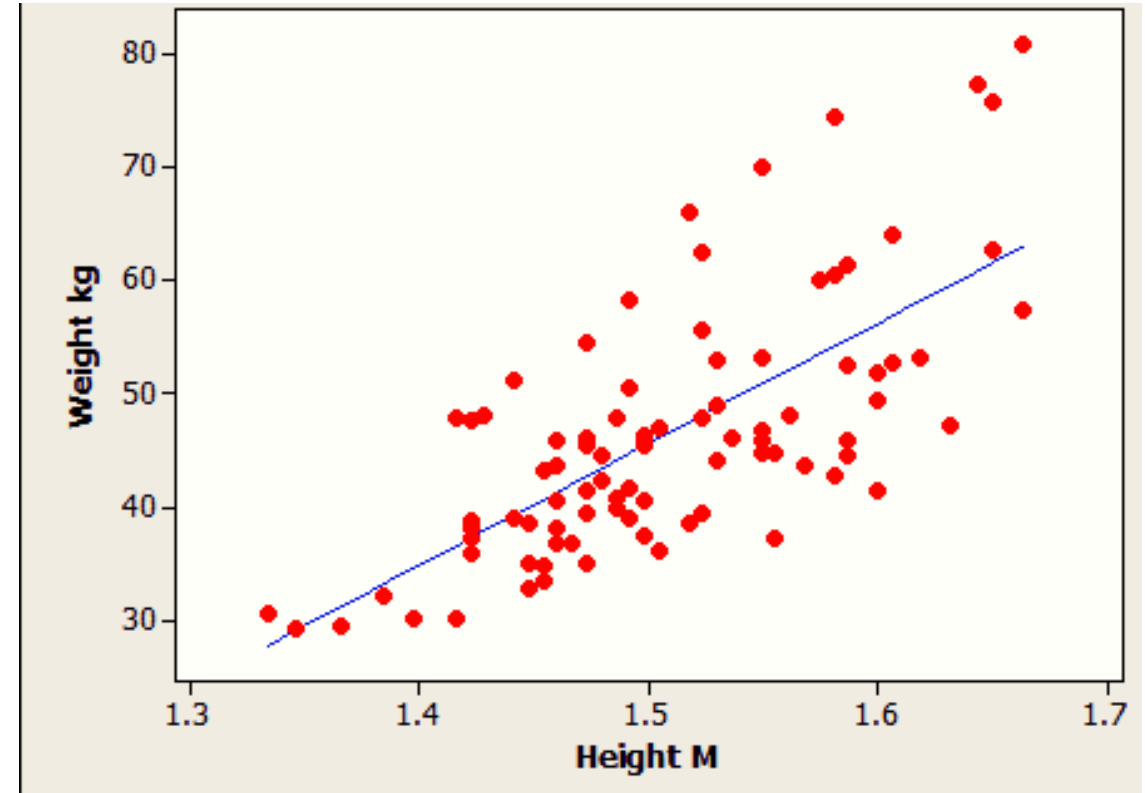
Slope

Independent variable

y intercept

Definition of "Regression coefficients":

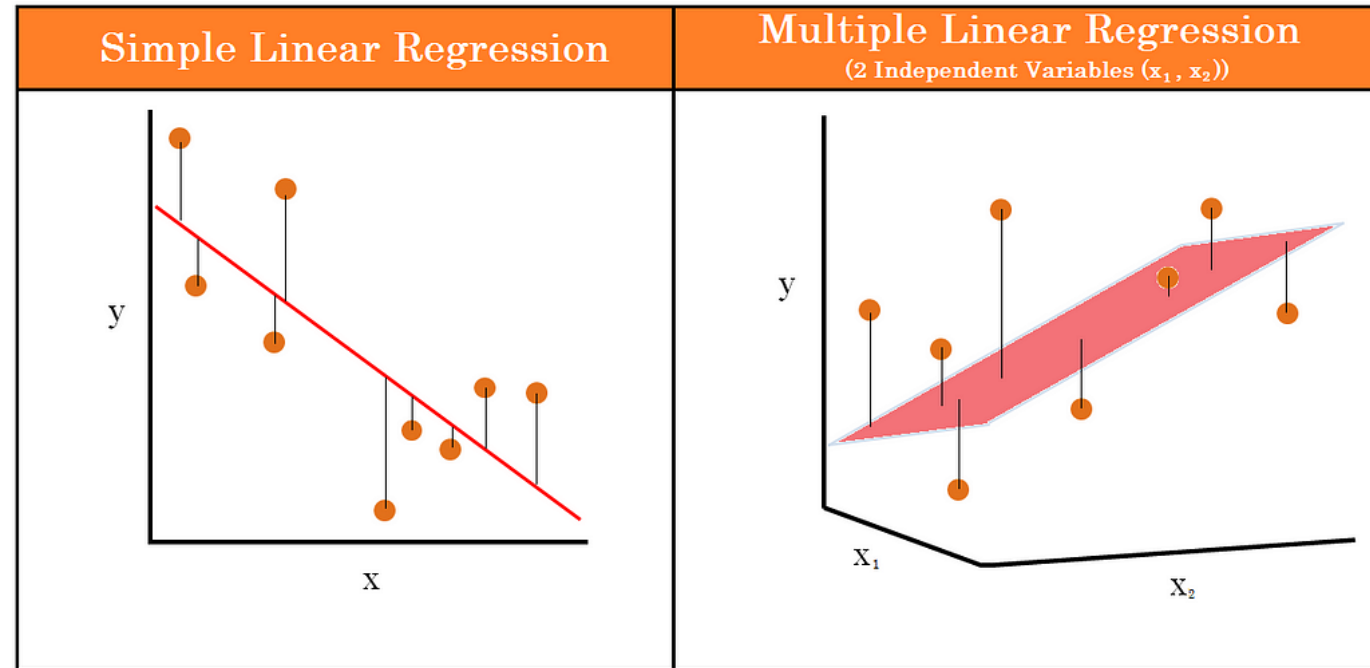
- **a** : point of intersection with the y-axis
- **b** : gradient of the straight line



Everything You Need to Know About Linear Regression

Tanvi Penmudy · Follow
Published in Analytics Vidhya · 9 min read · Jan 8, 2021

[Source](#)



Simple Linear
Regression

$$\hat{y} = b \cdot x + a$$



Multiple Linear
Regression

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + b_k \cdot x_k + a$$

Many examples will include multiple independent variables



[Source](#)

Agriculture example

	▽ metric	▽ metric	▽ metric	▽ metric
Case	Fertilizer Use (kg/l	Rainfall (mm)	Average Temperat	Crop Yield (tons/h
1	100	500	22	3.2
2	120	550	21.5	3.8
3	90	450	23	2.9
4	150	600	20	4.2
5	110	520	22.5	3.5
6	80	480	24	2.7
7	140	580	21	4
8	130	530	22.8	3.9
9	70	450	23.5	2.5
10	125	560	21.8	3.7
11	105	510	22.2	3.3
12	90	470	23.8	2.8
13	135	600	20.5	4.1
14	115	540	22	3.6

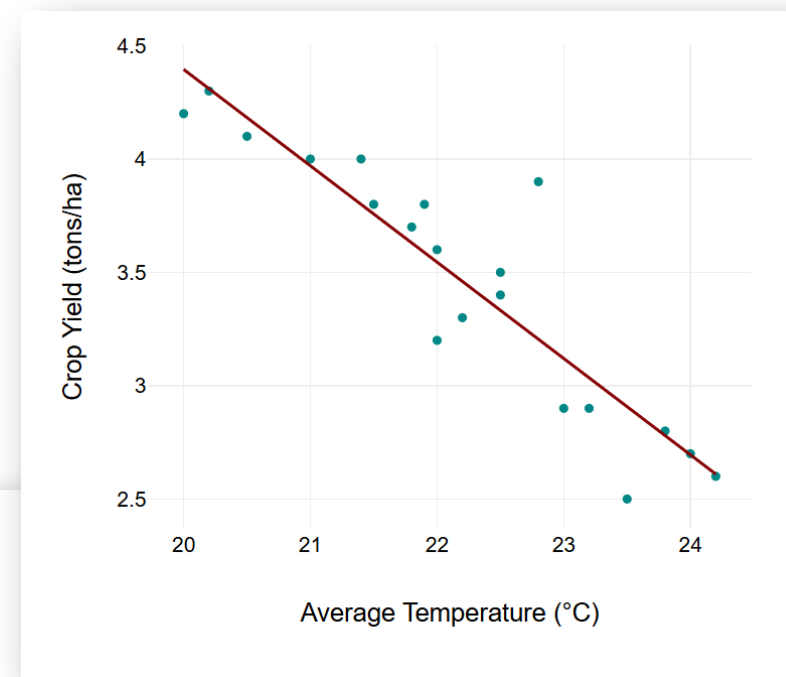
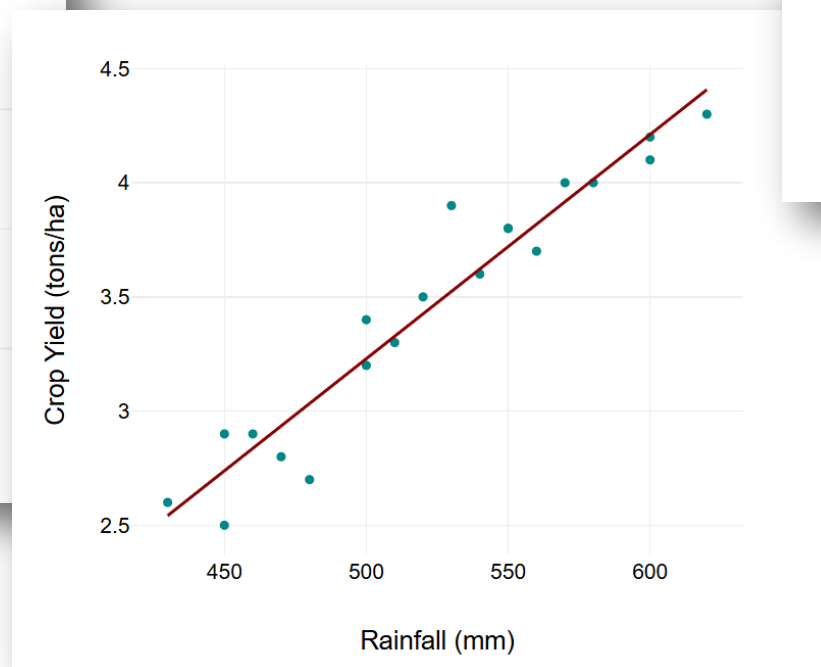
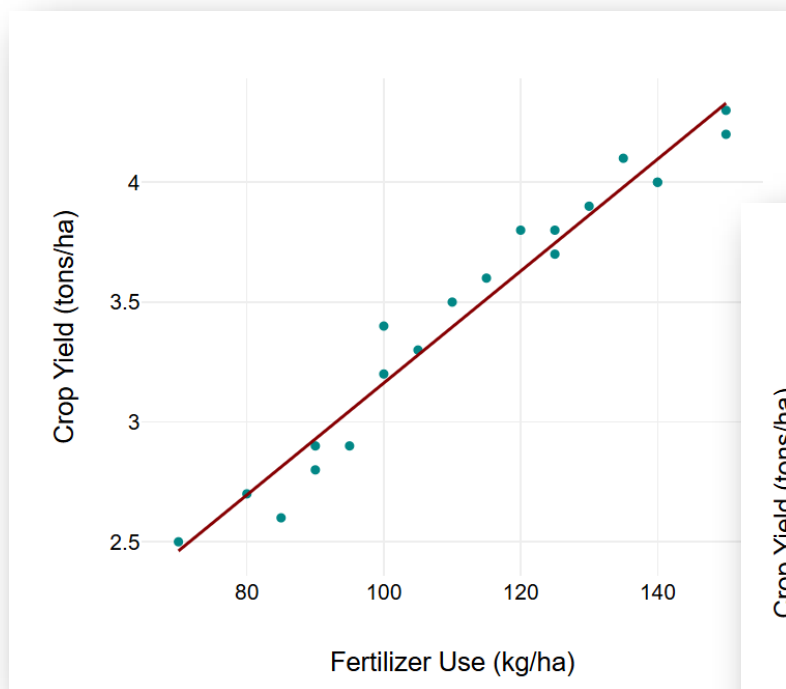
Multiple Linear Regression

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + b_k \cdot x_k + a$$



Exploring simple regression with
each feature independently

[Source](#)





[Source](#)

Medical example

	▽ metric	▽ metric	▽ metric	▽ metric
Case	Blood Pressure	Age	Weight	Cholesterol
1	78	58	76	153
2	91	48	70	234
3	79	34	52	195
4	93	62	99	190
5	85	27	62	243
6	90	40	55	223
7	92	58	81	183
8	73	38	69	207
9	93	42	99	202
10	86	30	73	246
11	92	30	93	234
12	95	43	84	225
13	86	55	73	204

Multiple Linear Regression

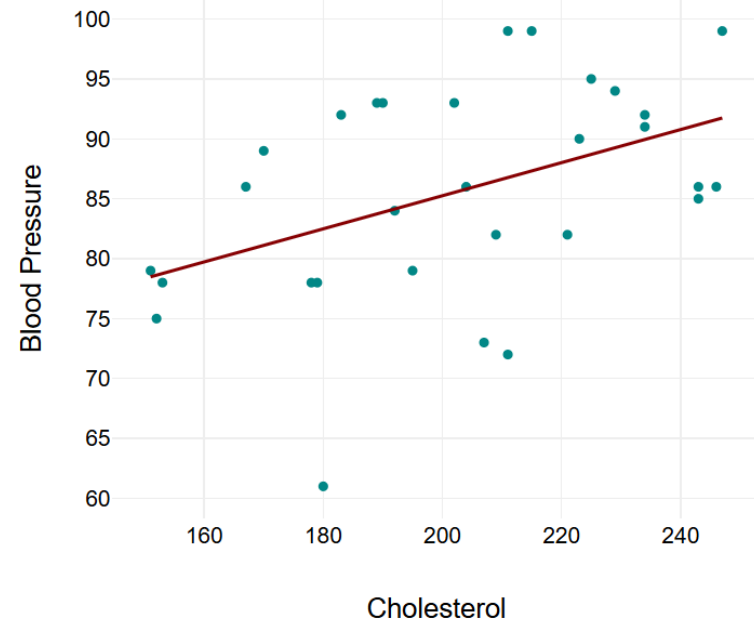
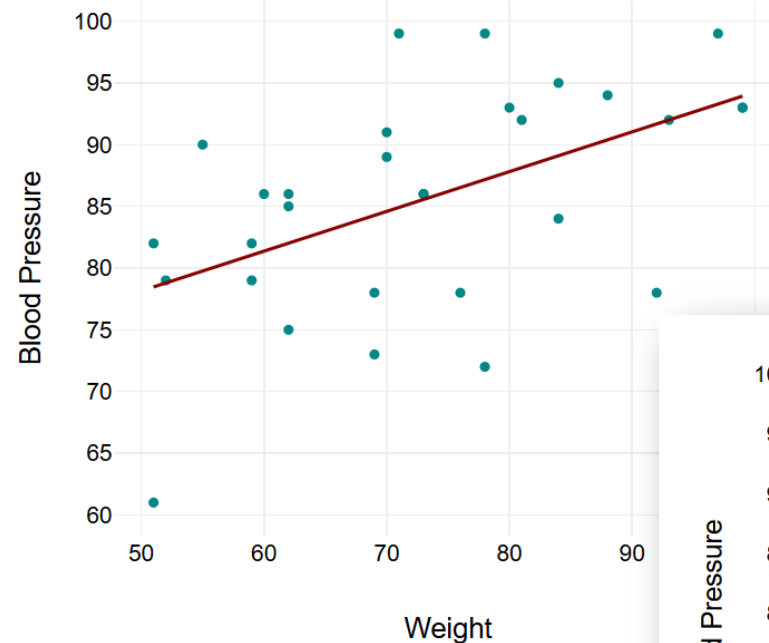
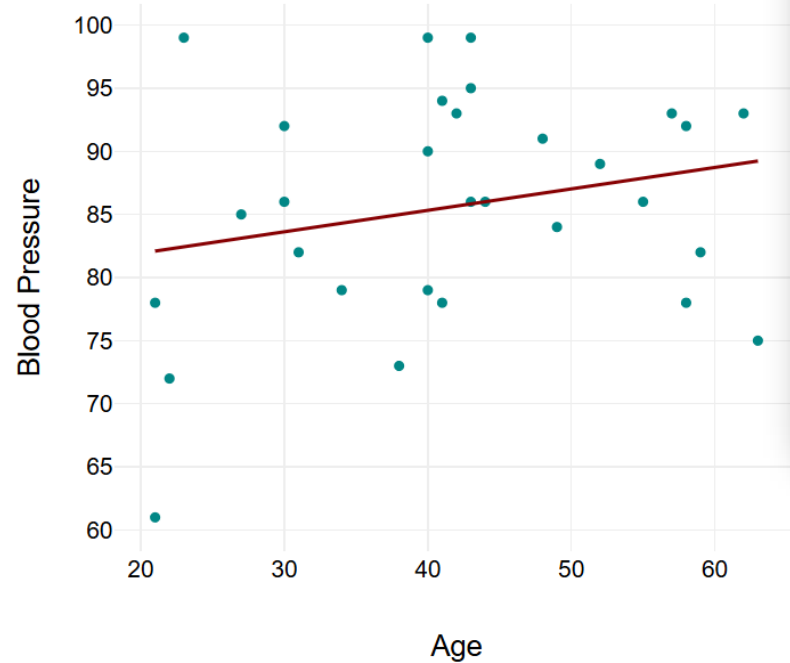
$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + b_k \cdot x_k + a$$



Exploring simple regression with
each feature independently

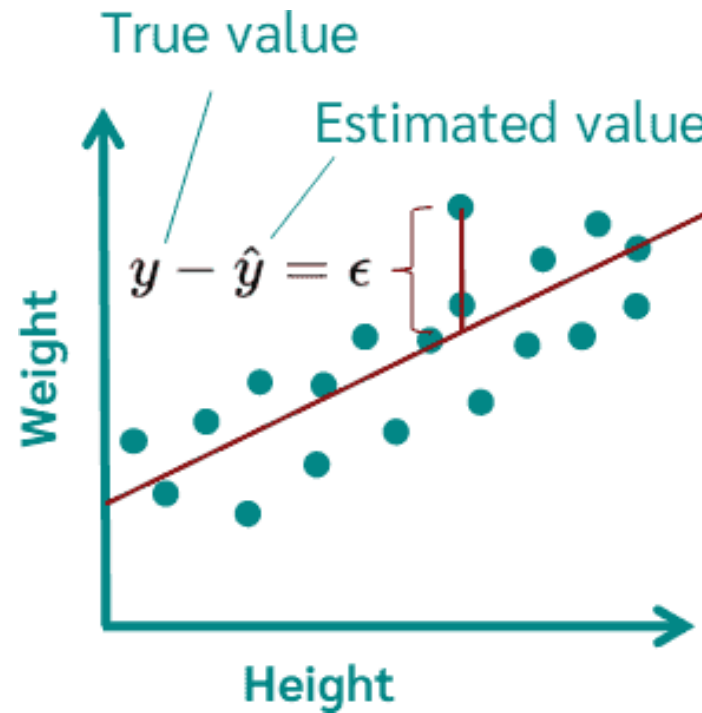
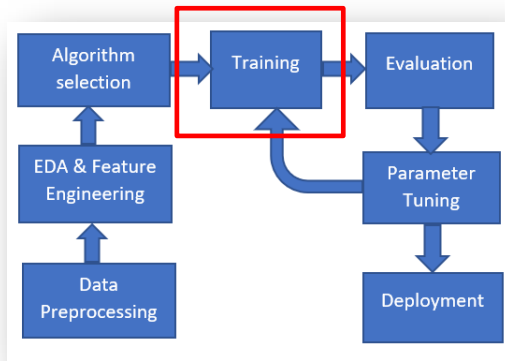
[Source](#)

Medical example



Model learning

Estimation of parameters



Parameter learning should consider the residuals over the full dataset

Error epsilon

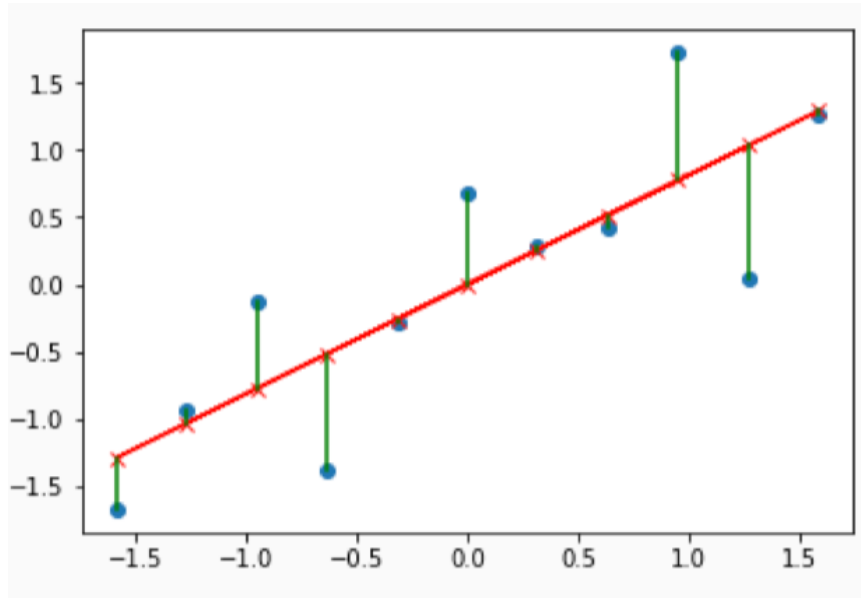
$$y = b \cdot x + a + \epsilon$$

To learn the model's parameters, we need a cost function to minimize

Calculate the **Residual Sum of Squares (RSS)**. *Residuals, or error terms*, are the difference between each **actual output** and the **predicted output**.

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

We want to learn the set of parameters that will minimize that function



Learning an LR Model with Gradient Descent

Learning the parameters can be done in an iterative manner, with an algorithm such as gradient descent

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + b_k \cdot x_k + a$$



$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + b_k \cdot x_k + b_0$$

Initialize parameters at random

Repeat:

For each example in training set:

- a. Predict $\hat{y}$ (forward pass)
- b. For each parameter b_i

- a. Calculate derivative of error

$$\frac{dE}{db_i}$$

- b. Update parameter

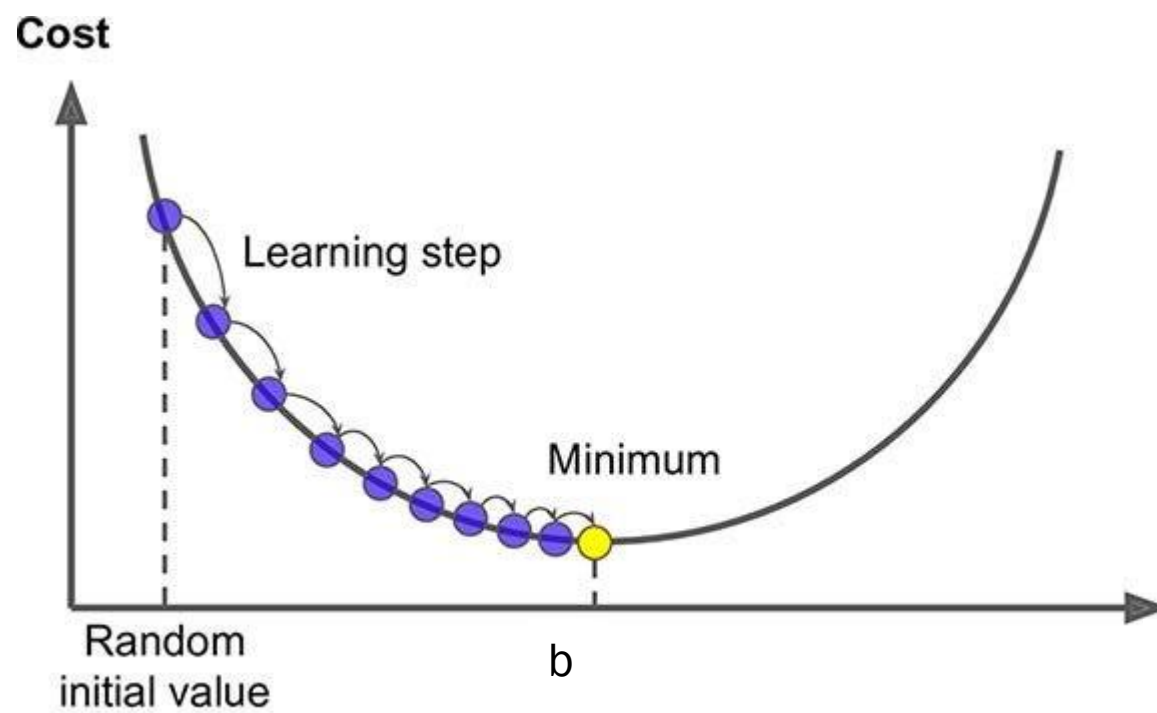
$$b_i^{(t)} = b_i^{(t-1)} - \alpha \frac{dE}{db_i}$$

Until "done"

What is « done »?

Gradient descent

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$





Lasso vs Ridge vs Elastic Net | ML

Last Updated : 07 Feb, 2025

[Source](#)

I encourage you to investigate further into variations of the cost function.

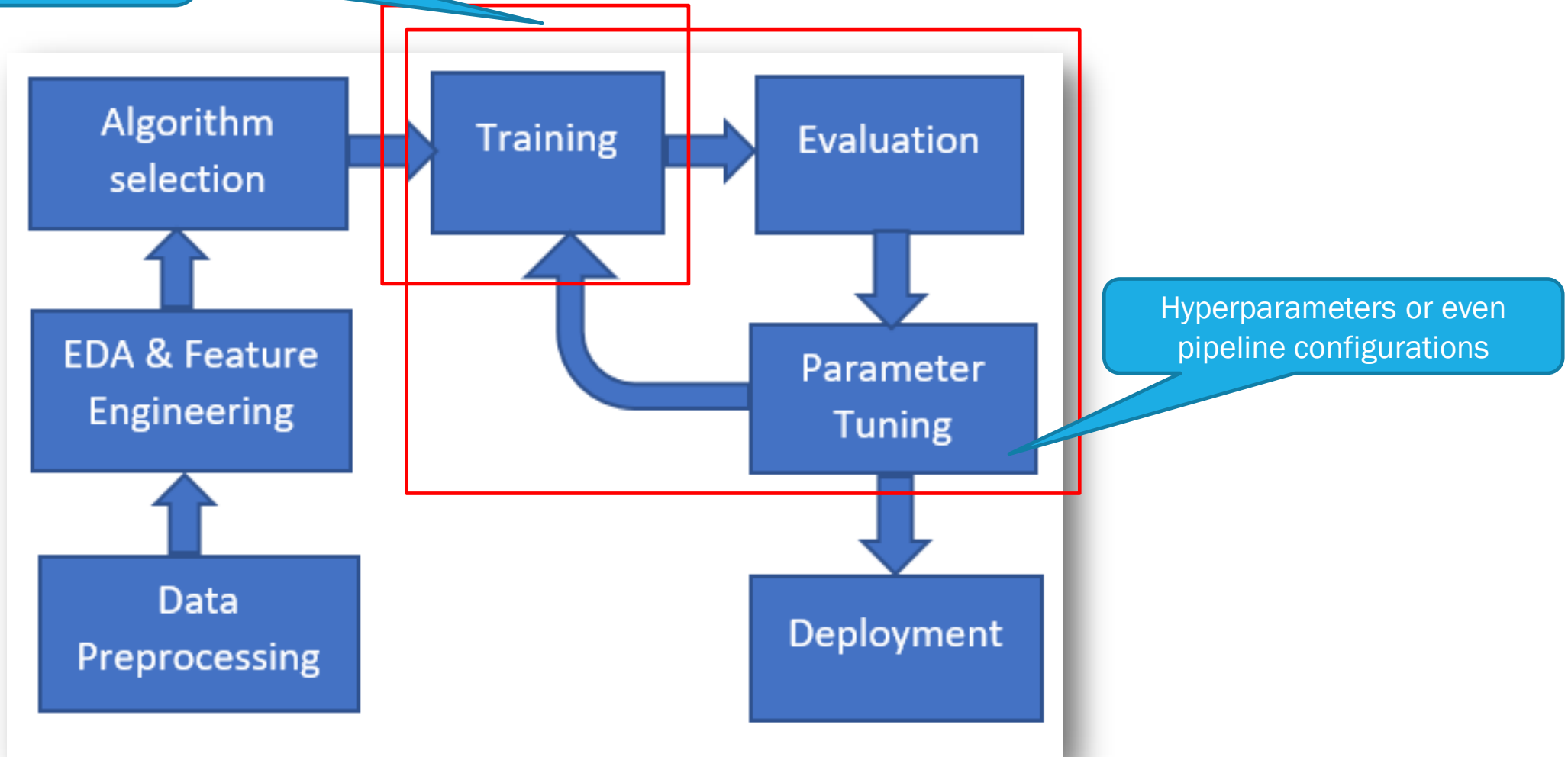
- Ridge Regression
- Lasso Regression
- Elastic Net

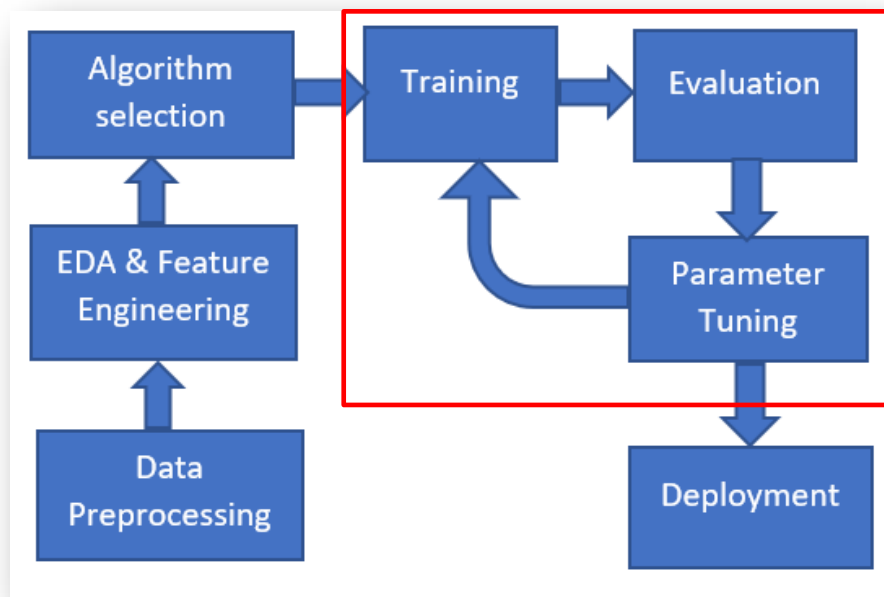
$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

Adding factors to keep the parameters small

$$\text{RidgeLoss} = \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n \beta_j^2$$

Once model is built, it can be evaluated and enhanced





Mean Square Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Use metrics to evaluate

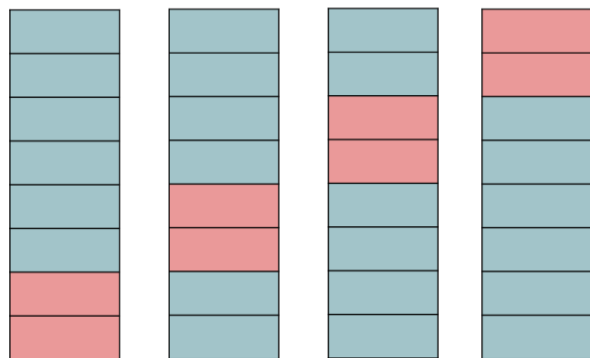
Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

R-squared

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

4-fold cross-validation



Assumptions

What Linear regression relies on



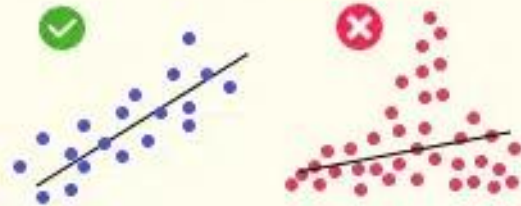
Source

I encourage you to read... I will only focus on the first one

Assumptions of Linear Regression

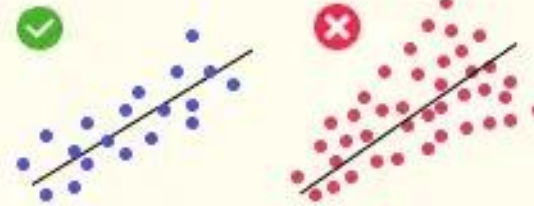
1. Linearity

(Linear relationship between Y and each X)



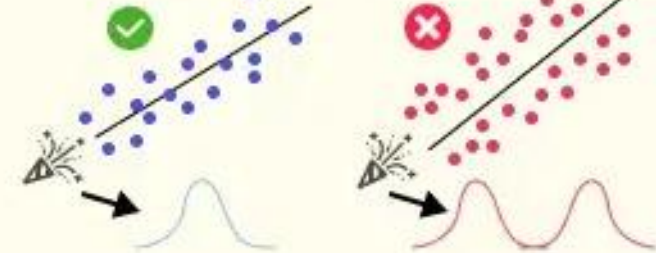
2. Homoscedasticity

(Equal variance)



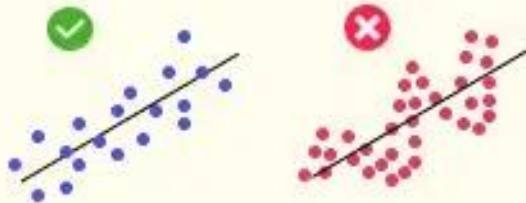
3. Multivariate Normality

(Normality of error distribution)



4. Independence

(of observations. Includes "no autocorrelation")



5. Lack of Multicollinearity

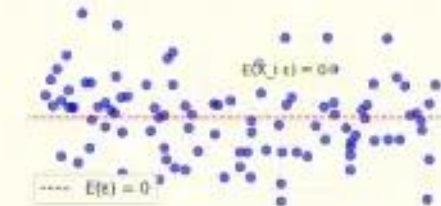
(Predictors are not correlated with each other)

$$X_1 + X_2$$

$$X_1 \sim X_2$$

6. Absence of endogeneity

(No correlation between predictors and errors.)

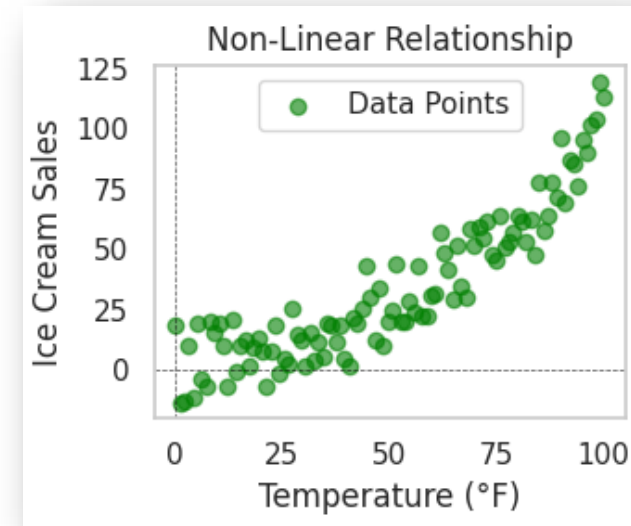
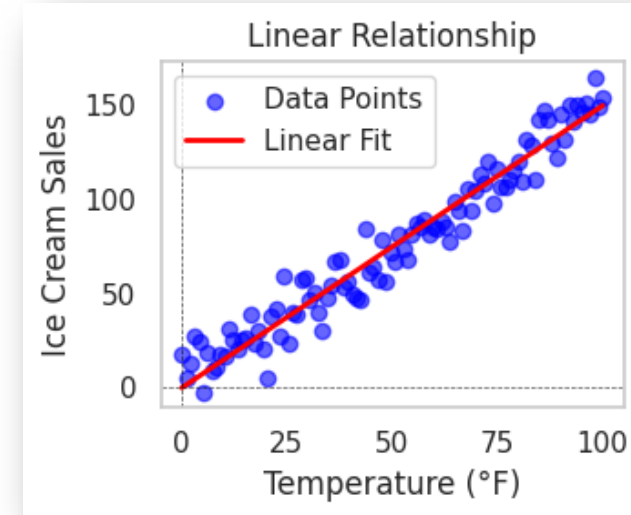


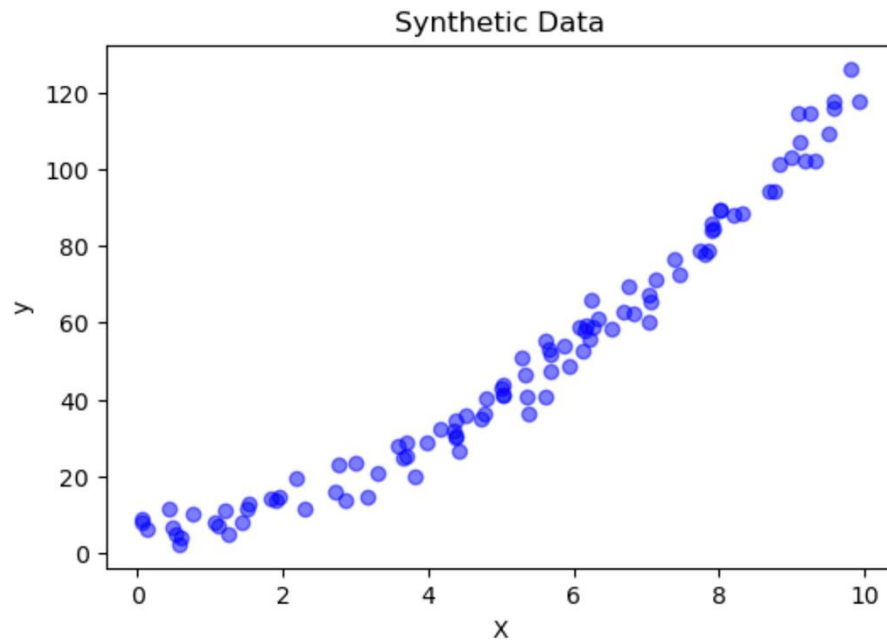
Linearity

The first and foremost assumption of linear regression is that the relationship between the predictor(s) and the response variable is linear.

This means that a ***change in the independent variable results in a proportional change in the dependent variable.***

This can be visually assessed using scatter plots or residual plots.

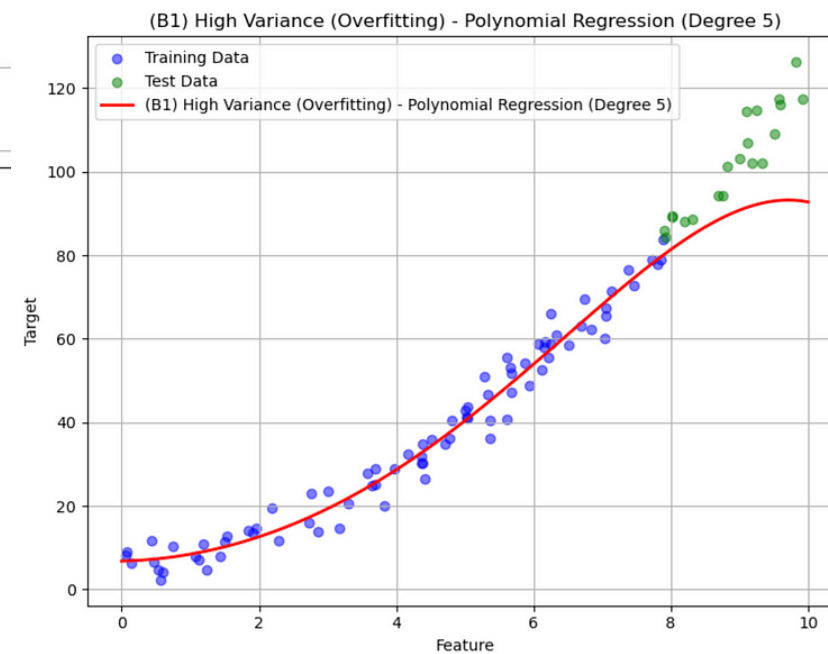
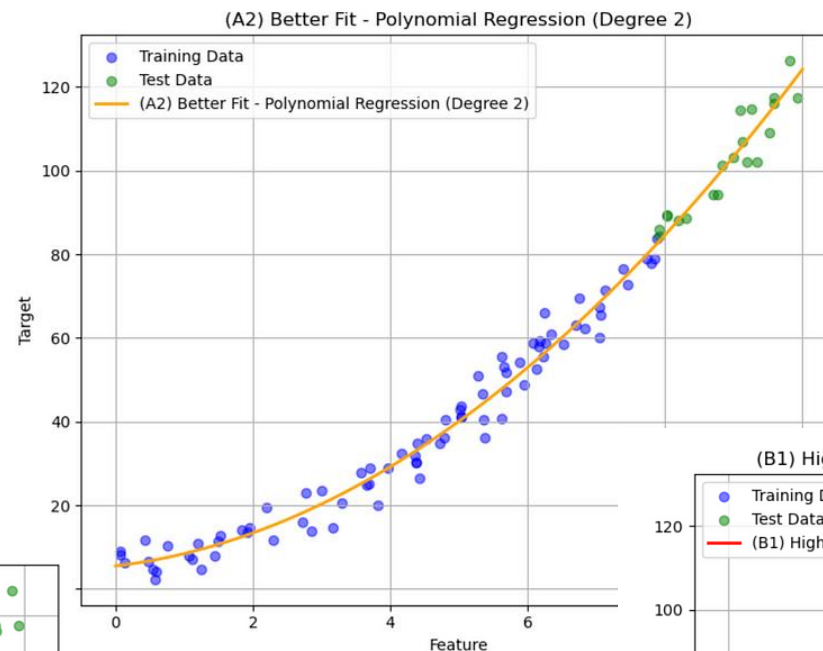
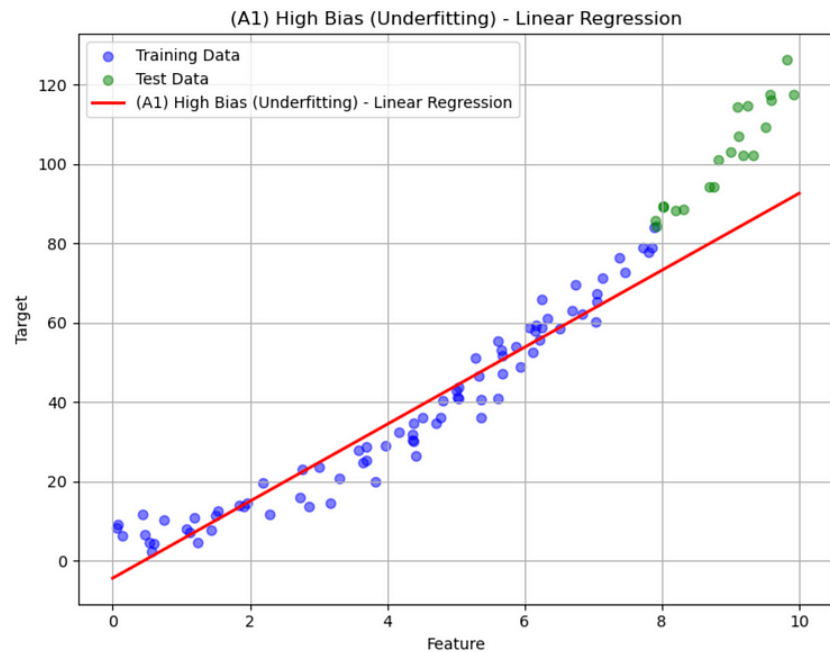




Synthetic data – expect a quadratic equation to fit and will try linear, quadratic and higher degree

Row	Model	Train Error	Test Error
1	Linear Regression	35.62	458.56
2	2nd Degree Polynomial Regression	13.70	18.96
3	5th Degree Polynomial Regression	12.98	254.48

Best « fit » 2nd degree polynomial



Linear regression

Prediction for Imputation

We already explored this model for imputation

Student	Math Score	Reading Score	Writing Score	Absences
A	85	88	90	5
B	72	75	78	3
C	?	80	85	7
D	90	92	94	2
E	65	70	68	4

$$\text{Math Score} = \beta_0 + \beta_1 \times \text{Reading Score} + \beta_2 \times \text{Writing Score} + \beta_3 \times \text{Absences}$$

- $\beta_0 = 50$ (intercept)
- $\beta_1 = 0.2$ (coefficient for **Reading Score**)
- $\beta_2 = 0.3$ (coefficient for **Writing Score**)
- $\beta_3 = -1.5$ (coefficient for **Absences**)

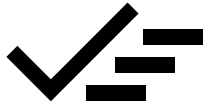
These parameters are learned for « best fit ».

$$\text{Math Score} = \beta_0 + \beta_1 \times \text{Reading Score} + \beta_2 \times \text{Writing Score} + \beta_3 \times \text{Absences}$$

Student	Math Score	Reading Score	Writing Score	Absences
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